Day 061 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Test Solutions by Substitution

Standards: 6.EE.B.5

Learning target: I can test a proposed solution by substitution.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Replace the variable with the proposed value, compute each side, and compare. A number is a solution only when it makes the original equation true.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 3

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01 / Learn

Replace the variable with the proposed value, compute each side, and compare. A number is a solution only when it makes the original equation true.

Worked example 1

Does x = 4 solve 3x + 6 = 18? Show substitution.

1. Replace the variable with the proposed value, compute each side, and compare.

2. A number is a solution only when it makes the original equation true.

3. 3 × 4 + 6 = 18

4. compare with 18.

5. State the result in the form requested: Yes; left side = 18.

Answer: Yes; left side = 18

Worked example 2

Select all values of x from the list that make x + 6 > 8 true. Explain why each rejected value fails.
A. 3
B. 1
C. 2
D. 4

1. Replace the variable with the proposed value, compute each side, and compare.

2. Subtract 6 conceptually: x must exceed 2.

3. Equality at 2 fails because the inequality is strict.

Answer: A, D - 3; 4

Day 061 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Does x = 6 solve 4x + 5 = 25? Show substitution.

02 / Constructed response

Does x = 5 solve 3x + 2 = 17? Show substitution.

03 / Constructed response

Does x = 4 solve 3x + 6 = 15? Show substitution.

04 / Constructed response

Does x = 4 solve 8x + 5 = 37? Show substitution.

05 / Constructed response

Does x = 5 solve 8x + 2 = 34? Show substitution.

06 / Constructed response

Does x = 3 solve 5x + 6 = 21? Show substitution.

07 / Constructed response

Does x = 4 solve 5x + 5 = 20? Show substitution.

08 / Multiple select

Select all values of x from the list that make x + 3 > 5 true. Explain why each rejected value fails.

A. 3
B. 4
C. 1
D. 2

09 / Multiple select

Select all values of x from the list that make x + 6 > 10 true. Explain why each rejected value fails.

A. 4
B. 6
C. 3
D. 5

10 / Constructed response

Does x = 2 solve 7x + 7 = 21? Show substitution.

Day 061 | Unit 05 | Answer key

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1. No; left side = 29

4 × 6 + 5 = 29; compare with 25.

6.EE.B.5 | Constructed response

2. Yes; left side = 17

3 × 5 + 2 = 17; compare with 17.

6.EE.B.5 | Constructed response

3. No; left side = 18

3 × 4 + 6 = 18; compare with 15.

6.EE.B.5 | Constructed response

4. Yes; left side = 37

8 × 4 + 5 = 37; compare with 37.

6.EE.B.5 | Constructed response

5. No; left side = 42

8 × 5 + 2 = 42; compare with 34.

6.EE.B.5 | Constructed response

6. Yes; left side = 21

5 × 3 + 6 = 21; compare with 21.

6.EE.B.5 | Constructed response

7. No; left side = 25

5 × 4 + 5 = 25; compare with 20.

6.EE.B.5 | Constructed response

8. A, B - 3; 4

Subtract 3 conceptually: x must exceed 2. Equality at 2 fails because the inequality is strict.

6.EE.B.5 | Multiple select

9. B, D - 6; 5

Subtract 6 conceptually: x must exceed 4. Equality at 4 fails because the inequality is strict.

6.EE.B.5 | Multiple select

10. Yes; left side = 21

7 × 2 + 7 = 21; compare with 21.

6.EE.B.5 | Constructed response